

Lab 13 - Temperature, Pressure, Density, Buoyancy

Exploratory Rationale

In this lab, you will investigate some physical and thermal properties of materials, including water. There are 4 stations. Groups will rotate 2 at a time through each station at fixed intervals. Each station will take between 30-45 minutes to complete. Please leave all equipment as you found it so that the next group can start each experiment from the beginning.

At each station, you will record measurements and/or observations in a [googlesheet](#). Use the College Physics 2e textbook and lecture notes to apply concepts from chapters 11, 13 and 14 to support explanations for the measurements and observations recorded. Apply these same concepts to answer the questions at the end of the lab, which you may do while you wait.

Objectives

- Collaborate with other students in problem-solving and laboratory activities.
- Follow laboratory procedures to complete experiments and write laboratory reports.
- Use tables, graphs, equations and diagrams to understand physical concepts.
- Relate experimental observations to the physical concepts from chapters 11, 13 & 14.

Safety Awareness

- Be careful and cautious while working with hot water and hot substances.
- Thermometers must be allowed to slowly adjust to changes in temperature. This is especially true going from boiling temperature to room temperature.
- Wear safety glasses and use gloves at stations as indicated.

Station 1 Heating & Cooling - part 1

Exploration: warm vs. cold tiles

Equipment: 2 identical tiles of different material and an infrared (IR) thermometer

Two tiles made of different materials are each at room temperature. Take turns so that each person in the group has a chance to place one hand on each tile simultaneously. Discuss and record your observations in the googlesheet. Is there a group consensus?

- 1) Which material feels colder? Describe what you feel in words.

The slate black material feels like cool metal, and the more gray one feels room temp or even slightly warmer.

- 2) Use the infrared thermometer to measure the temperature of each tile. How do the measured temperatures compare to room temperature? Are the temperatures of the tiles roughly the same or different?

Other black one: 21.4 C

Slate black one: 21.4 C

Room temp: 20.4 C

- 3) Read sections 14.1 & 14.4 of College Physics 2e and define temperature and heat here (include the units of each quantity):

Temperature = measure of the average kinetic energy of the particles (atoms or molecules) within a substance, indicating how hot or cold it is

Heat = Heat is the spontaneous transfer of energy due to a temperature difference. Heat is a form of energy, it has the SI unit of *joule* (J). The *calorie* (cal) is a common unit of energy, defined as the energy needed to change the temperature of 1.00 g of water by 1.00°C

- 4) Use the concepts in the textbook (or reference a reputable physics website) to explain in physics terms why the tiles feel different at room temperature.

The tiles feel different at room temperature because the “colder” tile has a higher thermal conductivity and therefore pulls heat away from our hands faster than the other tile. The same concept applied to the ice cubes.

- 5) **Which material will a cube of ice melt faster on? Record your predictions and explain your reasoning here.**

The cooler one would melt and ice cube faster. I have no clue why, but I saw Bill Nye do it once. This must be because it has better conduction of heat and therefore can add heat to the ice cube faster.

At the end of lab...

- 6) After every group has tested the tiles, the instructor will place an ice cube on each tile to test your predictions.

Which ice cube melted faster? Do your observations agree with your predictions?

The ice cube on the cooler slate block melted faster. This matched out predictions of what would happen.

Station 2 Heating & Cooling - part 2 (use gloves or hot pads)

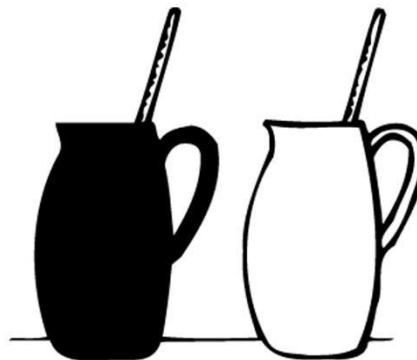
Exploration: Dark objects versus light objects

We know from life experience that black objects and clothing heat up faster in the sunlight than white objects and clothing. But have you ever thought about which color cools the fastest?

Two cans of the same size and shape are at this station. One is painted black and the other white for comparison.

Equipment:

2 identical cans of different color
1 pasco thermometer
1 digital thermometer
Capstone laptop
Electric kettle
IR camera

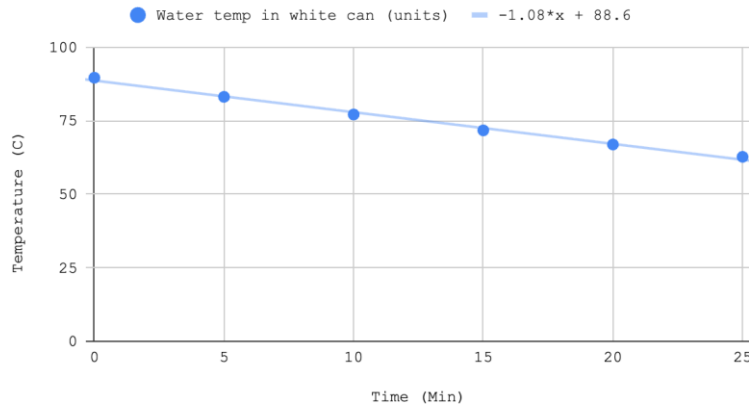


Instructions (use caution and plan timing):

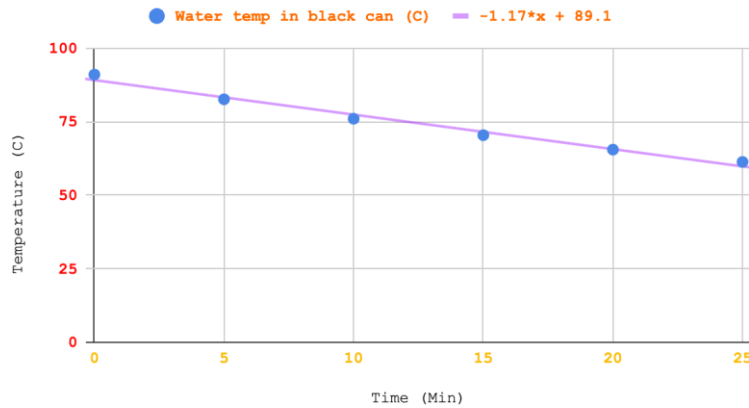
1. Bring 500 mL of water to boil (100°C) in an electric kettle.
2. Place both cans in the sink
3. Setup Capstone to record the temperature inside one can using the Pasco thermometer and a Temp vs. time graph.
4. Turn on a digital thermometer to use for the second can.
5. Carefully pour equal amounts of boiling water into each can and plug the cans with a stopper.
6. Insert a thermometer into each stopper and start recording with capstone. When the temperature reaches maximum (close to 100°C), this is your $t = 0$ min data point.

7. Record the temperatures of the water in the googlesheet, and repeat every 5 minutes for 20-30 minutes.
8. Create one scatter plot that shows temperature v. time for each can. Be sure to include a legend.

Time vs Temperature of White Can



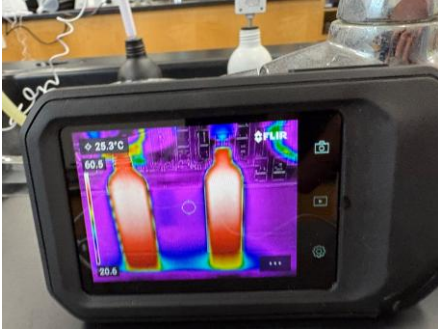
Time vs Temperature of Black Can



- 1) Based on your measurements, which can cooled down more quickly? Explain.

Based on the measurements the black can cooled down faster.

- 2) View the cans through the infrared camera. Sketch and describe what you see. Which can is cooler than the other?



Based on the data that we got the can on the left (black)

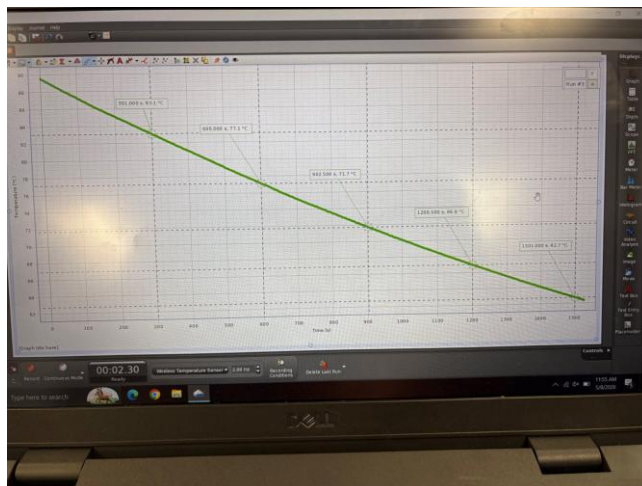
- 3) Carefully touch each can near the spout. Do you feel a difference in temperature?
 The black one feel more cooler because absorbs the heat, but because the white one reflect it the surface feels warmer

- 4) Use the concepts in the textbook (or reference a reputable physics website) to explain in physics terms why the water temperature decreases at different rates in the cans.

Darker surfaces emit heat faster than lighter surfaces, so the darker one can lose heat faster than the lighter one.

- 5) Insert a screenshot of your capstone graph here and describe how it compares to the scatter plot.

Our trend line for the scatter plot follows the same linear path as the capstone graph, which shows the linear relationship in how heat disperses from the can.



Station 3: Heating & Cooling - part 3 (use safety glasses and thermal gloves)

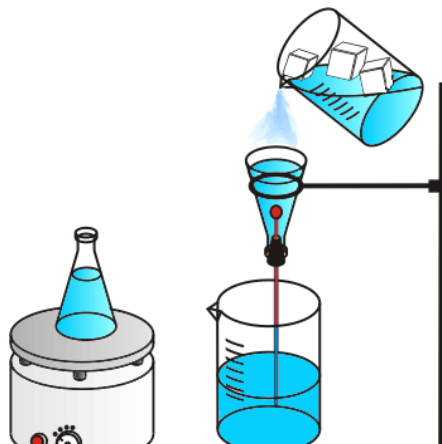
Exploration: the boiling point of water

The boiling temperature of a liquid depends on the ambient pressure and can be determined from a phase diagram, such as Fig 13.29 in section 13.5 of College Physics 2e.

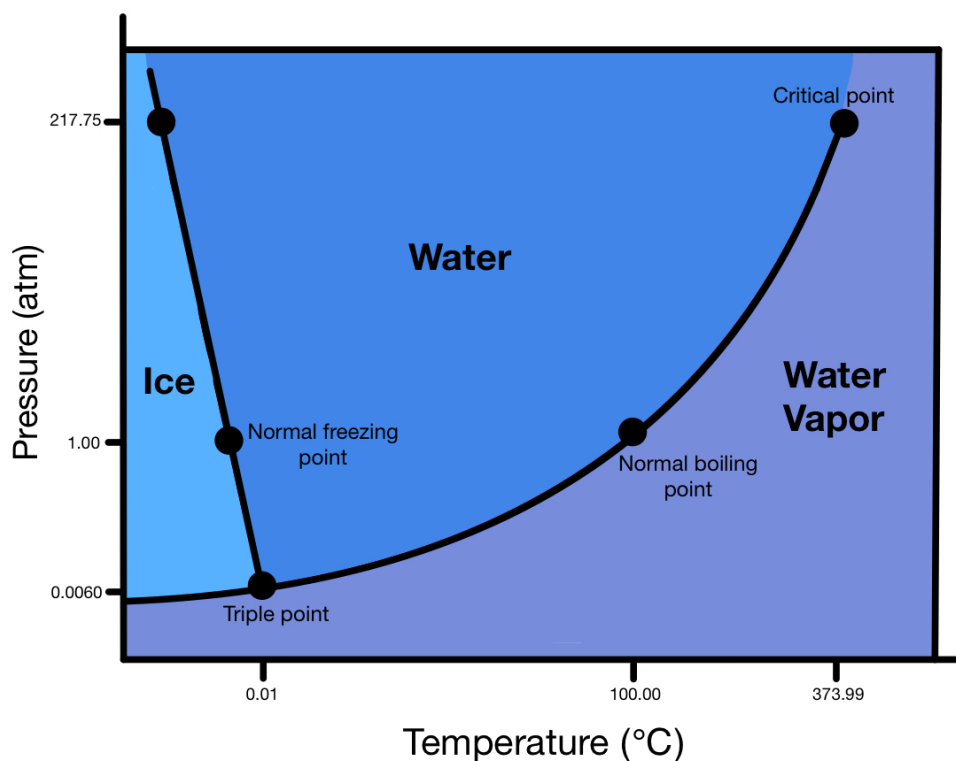
In this exploration, you will observe how pressure affects the boiling point of water.

Equipment

- Erlenmeyer flask
- one stopper
- electric kettle
- two beakers
- ring stand
- clamp
- hot plate
- ice



Phase Diagram of Water



Instructions (steps 1-5 are preparation for the experiment):

1. Warm the empty Erlenmeyer flask on the hot plate (to warm the air inside)
2. Bring 500 mL of water to boil (100°C) in an electric kettle.
3. Fill one beaker half full with tap water and the other with ice water.
4. Fill the flask with boiling water. Secure the stopper and gently shake the flask for 3 to 5 seconds over the sink. (Use gloves to hold the flask and keep your thumb on the stopper).
5. Unplug the flask and pour out all the water into the sink.
6. Fill the flask about 75% full with boiling water (leave approximately 25% empty).
7. Secure the stopper in the mouth of the flask.
8. Carefully position the flask upside down in the ring stand in the sink.

9. Pour the beaker of tap water over the bottom of the flask and **record your observations of the water inside in the googlesheet. Does the water start boiling?**

It does start boiling. When pouring the water over the upside down flask it cools the top of the flask leading to a change in pressure.

10. When tap water no longer causes the water to boil, pour the beaker of ice water over the flask and **record your observations. Does the water start boiling again?**

We didn't see the water boil again but we expected the same result to occur as with the room temp water as the air around the water cools and the pressure changes.

- 1) **Read section 13.5 of College Physics 2e. Compare Fig 13.29 with the Phase Diagram of Water above. Is one figure easier to understand than the other? If so, why?**

Yes, the figure 13.29 is easier to understand because it is very self explanatory. It has only three axes, and there are less lines to try to understand. There are only three lines and 3 areas where data must be interpreted.

- 2) **Use the phase diagram and the concepts in the textbook to explain in physics terms why the water in this activity boiled when cooled.**

The heated water was kept in a sealed flask with warmed air. When the room temp water was poured over it the air was cooled so it contracted and reduced atmospheric pressure on the water. This means the vapor pressure required for water to boil is lowered as well because it has to be equal to atmospheric pressure. Once it had adjusted to that ice water was poured over repeating the process using the same logic.

- 3) Would the water have boiled if the cooling flask were open to the atmosphere?
Explain.

The water would have not boiled if exposed to the atmospheric pressure because with the flask sealed, the system is closed so during the cooling process, there is nowhere for the air to escape.

Station 4: Density & Buoyancy (derive equations first)

Exploration: determine the density of unknown materials

In this experiment, you will measure the density of 4 unknown metals in two different ways. You will compare these densities to one another and to the known densities of metals in order to identify the unknown metals and to account for experimental discrepancies.

Equipment:

4 identical masses of different material
Pasco force sensor
Capstone laptop
String
Beaker
Calipers
Mass scale

Using sections 11.2 and 11.7 of College Physics 2e and the lecture notes and slides to...

1). Write an equation for the density of an object ρ in terms of its mass m , and volume V .

$$\rho = m/V$$

2). Write an equation for the weight of the object as read by a spring scale if the object is attached by a light string to the force sensor and hanging in air Label it W_g or W_{air} .

$$W_g = mg = \rho V g$$

3). Derive an equation for the apparent weight of the object as read by the spring scale if the object is attached by a light string to the force sensor and submerged in fluid. Label it $W_{apparent}$ or W_{fluid} . Leave your answer in terms of the buoyant force F_B , the mass of the object m , and g .

$$W_{apparent} = W_{in_fluid} = mg - F_B$$

4). Write the equation for the buoyant force F_B acting on the object in a fluid in terms of its volume V , g , and the density of the fluid ρ_f .

$$F_{Buoyancy} = \rho_f V g$$

5). Using your answer to the previous questions, write an equation for the density of the object ρ in terms of W_f , ρ_f , g , V only.

$$\rho = W_f / Vg + \rho_f$$

Instructions:

1. Setup Capstone with the Pasco force sensor.
2. Open a new graph and open a new table to record force measurements.
3. Use a light string to connect the force sensor to one of the masses.
4. Record the force on the sensor (zero the sensor while there is no tension in the string).
5. Measure the weight of the mass in air W_g with tension in the string.
6. Measure the apparent weight of the mass in water W_{fluid} with tension in the string.
7. Record your measurements in the googlesheet.
6. Calculate the density of the mass from the equation you derived using a cell formula.
7. Repeat steps 4-6 for the remaining masses.
8. Use the table in section 11.2 of College Physics 2e to identify the material of each mass.

If time allows ... NO TIME

9. Use another method to determine the density of the blocks. Show your work.

10. Calculate the %difference between your measurements. Show your work.

11. Identify possible sources of error and list them.

Applicaiton Questions (answer 1 question after each station)

Use the concepts from the lab (as well as the textbook or other reputable physics website) to answer the following questions in physics terms and in complete sentences:

1. Assuming water in a swimming pool is at the same temperature as the air, will you feel cooler immediately after climbing out of the pool or after you have dried off?

After leaving the pool, water remains in your skin until it evaporates, which requires energy, this energy is taken from your skin causing the temperature to decrease. Even if the water and the air have the same temp. Once you dry, the evaporation stops, so you are no losing heat as rapidly and you don't feel too cool

2. If you were in a very hot desert standing under the shade of a palm tree and you wanted to cool down, which would be better to wear, white clothes or black clothes?

It would be better to wear white clothes because light colors reflect sunlight while darker clothes absorb sunlight and in turn absorb heat.

3. In addition to the normal cooking directions printed on the back of a box of rice, there are also "high-altitude directions." The only difference is that the "high-altitude directions" suggest increasing the cooking time and using a greater volume of boiling water in which to cook the rice. Why should the directions depend on the altitude in this way?

At higher altitudes, atmospheric pressure is lower, causing water to boil at a temperature below 100°C. Because the boiling water is at a lower temperature, the rice cooks more slowly, as it requires a sufficient amount of thermal energy.

4. How did Archimedes use water displacement to determine whether the crown of King Hiero II of Syracuse was made of pure gold?

A real crown made of pure gold would be very dense, thus the crown would quickly sink if it was real, but if it was made of a non gold metal it would float or sink very slowly because it is less dense

